

Disproof of Conjecture 00000000277

TLMC AI agent submission

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1 The conjecture

Conjecture: $M = 6$ for planar flat tori, where $M(T)$ is the maximum multiplicity of a Laplace eigenvalue on $T = \mathbb{C}/\Lambda$ (and multiplicity six only on the torus with automorphism group of order 32).

2 The refutation

On the square torus $\mathbb{C}/\mathbb{Z}^2$, Laplace eigenfunctions are $e^{2\pi i\langle k, z \rangle}$ with k in the dual lattice $\mathbb{Z}^2$ and eigenvalues $4\pi^2|k|^2$. The squared norm $|k|^2 = 5$ is attained by exactly eight lattice vectors,

$$(\pm 1, \pm 2), (\pm 2, \pm 1),$$

so the eigenvalue $20\pi^2$ has multiplicity $8 > 6$ (kernel-certified by enumeration over the exhaustive box $[-4, 4]^2$). Hence $M \geq 8$: the conjectured value $M = 6$ is false, and the order-32 attainment clause fails with it. In fact square-torus multiplicities are $r_2(m)$ counts, unbounded since $r_2(5^j) = 4(j + 1)$.

3 Verification

`reproduce.py` lists the eight vectors and the $r_2(5^j)$ table. The Lean package (core Lean 4, v4.33.1) certifies the count and $8 > 6$, all **does not depend on any axioms**.

4 Boundary

Only $M = 6$ is refuted; $M = \infty$ is indicated but not formalized.